



Department of Mathematical & Physical Sciences

EAST WEST UNIVERSITY

Aftabnagar, Dhaka 1212, Bangladesh

Course Outline of Course PHY 209 (Engineering Physics-II)

Semester: Summer 2026; Sections: 2 & 3

Course Instructor: Name - Md. Mahfuzul Haque, Code - MMHQ

Course Information

Course Code : PHY 209

Course Title : Engineering Physics – II

Credit Hours : 3

Class Time	Section	Day	Time	Room No.
	2	MW	08:30AM - 10:00AM	AB1-802
	3	SR	03.10PM - 04.40PM	AB1-801

Office Hour :

Days	Office Hours
Sun (S)	01.30 PM-03.00 PM
Mon (M)	10.00AM-11.30AM
Tues (T)	01.30 PM-04.00PM
Wed (W)	03.15PM-04.45PM
Thurs (R)	01.30PM-03.00PM 04.45PM-06.15PM

Make-up Schedule : Makeup classes will be suitably arranged.

Instructor Information

Course Instructor : Md. Mahfuzul Haque

E-mail : mahfuzul.haque@ewubd.edu

Office : **Room # 705, Building AB1**

Course Objective

The principal objectives for this course are for you to learn the fundamental concepts, principles, and theories of foundations of quantum physics and to develop the ability to solve problems. Lectures are structured to help you understand the conceptual basis of modern quantum physics and examples are designed to re-enforce those concepts. This course will give the students the necessary insights and skills required to understand the physics behind the electronic and optoelectronic properties of solid-state materials in general and of semiconductor structures in particular. The students will be prepared for courses concerning electronic components, lasers, optical materials, optoelectronic systems, non-linear and quantum optics, Quantum computing, etc.

Learning Outcomes

After completing the course, the students will be able to:

- Explain qualitatively and quantitatively the role of photons in understanding phenomena such as the photoelectric effect, X-rays and Compton Effect.
- Gain an understanding to discuss and interpret experiments displaying wavelike behavior of matter, and how this motivates the need to replace classical mechanics by a wave equation of motion for matter (the Schrödinger equation).
- Understand the central concepts and principles of quantum mechanics: the Schrödinger equation, the wave function and its physical interpretation, stationary and non-stationary states, time evolution and expectation values.
- Demonstrate an understanding of the significance of operators and eigenvalue problems in quantum

mechanics.

- Independently solve the Schrödinger equation for simple one-dimensional systems – the square well, harmonic oscillator, potential barrier etc.
- Solve simple problems in two- and three-dimensions in various coordinate systems, e.g. by using separation of variables in the Schrödinger equation.
- Develop a knowledge and understanding of the concept that quantum states live in a vector space.

Course Contents/Descriptions

1. Emergence of Quantum Mechanics

Black body radiation, photoelectric effect, Compton Effect; de Broglie hypothesis and wave particle duality.

2. Wave Mechanics

Wave packets and the Uncertainty relations; the Schrödinger equation; Axioms of wave mechanics; the probability interpretation; Stationary states; eigenfunctions and eigenvalues; quantum probability; equation of motion in Schrödinger picture; Ehrenfest's theorem; the parity operator; the uncertainty relation.

3. One-Dimensional Potentials

The free particle; the infinite square well; potential barriers and scattering states; the single step potential barrier; Simple harmonic oscillator.

4. Three Dimensional Potentials

Three-dimensional infinite potential well; Schrödinger equation in spherical polar coordinates; separation into angular and radial components; the rotor; the hydrogen atom.

5. Formulation of Quantum Theory

Hilbert space and linear operators; Dirac notation; eigenstates and eigenvalues; Hermitian operators and observables; matrix representation of operators; unitary transformation; commuting operators; postulates of quantum mechanics.

Extra: Quantum Computing (If time supports)

Introduction to quantum computer; quantum bit (qubit); single qubits; the state space of a single qubit; multiple qubit systems; quantum state transformation – quantum logic gates; simple search algorithms.

Text Book:

1. Concepts of Modern Physics, Arthur Beiser
2. Modern Physics, Kenneth S. Krane
3. Quantum Mechanics: Concepts and Applications, Nouredine Zettili.

Reference Book:

1. Introduction to Quantum Mechanics, David Griffiths.

Assessment Tools:

Assessment tools include **Class Tests (Short Quizzes)**, **Presentations/Assignment**, and **Exams**. The Class Tests are about 20-30 minutes in class and the Mid Term Exam and Final Exam are 60 - 80 minutes duration in class. Class Test and Exam dates will be announced in the Class.

Assessment/Evaluation/Grading Policy:

The relative contributions of class tests, presentations and exams are as follows:

Test/Exam	% of Marks
Class Tests (Quiz)	20
Assignment/Presentation	10
Mid Term Exam	30
Final Exam	40

The University Grading Scheme is the following

Range of Marks (%)	Letter Grade	Grade Point	Range of Marks (%)	Letter Grade	Grade Point	Range of Marks (%)	Letter Grade	Grade Point
80 – 100	A+	4.00	65 – 69	B+	3.25	50 – 54	C+	2.50
75 – 79	A	3.75	60 – 64	B	3.00	45 – 49	C	2.25
70 – 74	A-	3.50	55 – 59	B-	2.75	40 – 44	D	2.00
						Less than 40	F	0.00

Essential Policy Information

1. There is zero tolerance for cheating at EWU. Students caught with cheat sheets in their possession, whether used or not used, &/or copying from cheat sheets, writings on the palm of hand, back of calculators, chairs or nearby walls, etc. would be treated as cheating in the exam hall. The only penalty for cheating is expulsion from EWU.
2. Regular Class attendance will be taken. The students are advised to attend classes regularly.
3. Makeup exam for mid-term and final may be arranged for compelling reasons. For the makeup exam, the student must apply with supporting documents.



(Md. Mahfuzul Haque)

Course Instructor